

AMAZON PRODUCT ANALYTICS USING DATABRICKS SQL AND POWER BI

End-to-End Data Engineering Project

PROJECT DOCUMENTATION

SUBMITTED BY

Varshini VB

ACADEMIC YEAR

2026

DECLARATION

I hereby declare that this project report titled "**Amazon Product Analytics Using Databricks SQL and Power BI**" is an original work carried out by me.

The work presented in this report has been completed as part of my learning in Data Engineering and Business Intelligence. The content of this report has not been copied from any previously submitted work except where proper acknowledgment has been provided.

The project demonstrates the complete implementation of an end-to-end Data Engineering solution using Databricks SQL, following the Medallion Architecture. The workflow includes data ingestion, data cleaning, transformation, ETL pipeline creation, validation, logging, analytical table creation, and interactive dashboard development using Power BI.

All SQL scripts, dashboards, and documentation included in this report were developed as part of this project.

Varshini V B
Signature

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who contributed directly or indirectly to the successful completion of this project.

I extend my heartfelt thanks to my trainers and mentors for providing continuous guidance, valuable suggestions, and technical knowledge throughout the development of this project.

I would also like to thank the Databricks platform for providing an excellent cloud-based environment to perform data engineering tasks efficiently. Additionally, I acknowledge Microsoft Power BI for enabling the creation of interactive dashboards and business intelligence reports.

Finally, I thank my family and friends for their encouragement and support throughout my learning journey.

TABLE OF CONTENTS

1. Project Overview	4
2. Problem Statement	5
3. Objectives	5
4. Technology Stack	6
5. Dataset Description	7
6. Project Architecture	8
7. Medallion Architecture	9
8. Data Ingestion	10
9. Data Cleaning	10
10. Exploratory Data Analysis (EDA)	11
11. Business SQL Analysis	12
12. Window Functions	13
13. Advanced SQL	14
14. Gold Layer	15
15. Data Validation	16
16. ETL Logging	17
17. ETL Pipeline	18
18. Power BI Dashboard	19
19. Business Insights	20
20. Challenges Faced	21
21. Future Enhancements	21
22. Conclusion	22

1. PROJECT OVERVIEW

Introduction

In today's digital marketplace, e-commerce platforms generate enormous volumes of product data containing information about pricing, discounts, customer ratings, reviews, and product categories. Organizations depend on this data to understand customer preferences, monitor product performance, optimize pricing strategies, and support business decision-making.

However, raw datasets obtained from online marketplaces are rarely ready for analysis. They often contain duplicate records, missing values, inconsistent formatting, and incorrect data types, making them unsuitable for direct reporting or business intelligence.

This project presents an end-to-end Data Engineering solution using Databricks SQL and Power BI to transform raw Amazon product data into meaningful business insights. The project follows the Medallion Architecture, a modern data engineering design pattern consisting of Bronze, Silver, and Gold layers.

The workflow begins by loading the raw CSV dataset into the Bronze layer without modification, ensuring that the original source data is preserved. The data is then processed within the Silver layer, where cleansing operations such as duplicate removal, null handling, data type conversion, and standardization are performed. Finally, business-focused analytical tables are created within the Gold layer to support reporting and dashboard creation.

To ensure reliability, the project also includes data validation procedures, ETL logging, and an automated ETL pipeline. The processed Gold layer data is integrated with Power BI to build interactive dashboards that provide valuable insights into product pricing, customer ratings, discount patterns, and category performance.

Project Highlights

- Designed an end-to-end Data Engineering workflow.
- Implemented the Medallion Architecture (Bronze, Silver, Gold).
- Performed data ingestion and transformation using Databricks SQL.
- Developed reusable SQL scripts for data cleaning and analytics.
- Built Gold layer tables optimized for reporting.
- Implemented data validation and ETL logging.
- Created an ETL pipeline for workflow orchestration.
- Developed interactive dashboards in Power BI.
- Managed the project using GitHub for version control and documentation.

2. PROBLEM STATEMENT

Modern organizations rely heavily on accurate and reliable data to support strategic business decisions. Amazon product datasets contain valuable information such as product names, categories, prices, discounts, ratings, and customer reviews. However, raw datasets obtained from external sources often contain inconsistencies that limit their usability for business reporting.

Major Issues Identified in Raw Data:

- Presence of duplicate product records.
- Missing values in important attributes.
- Inconsistent formatting of price and discount fields.
- Incorrect or mixed data types.
- Invalid or incomplete rating information.
- Lack of structured analytical tables for reporting.

Without proper data engineering processes, these issues reduce data quality and negatively impact business intelligence reports. Therefore, there is a need to design an automated ETL workflow capable of ingesting raw data, performing cleansing and validation, transforming the data into analytical models, and generating dashboards that provide meaningful business insights.

3. OBJECTIVES

The primary objective of this project is to build a scalable and structured Data Engineering solution capable of transforming raw Amazon product data into analytics-ready datasets. The specific objectives are:

- To ingest raw Amazon product data into the Bronze layer.
- To implement data cleaning and transformation in the Silver layer.
- To design analytical Gold layer tables for reporting.
- To perform exploratory data analysis using SQL.
- To answer business questions using advanced SQL techniques.
- To implement data validation rules for ensuring data quality.
- To maintain ETL execution logs for monitoring purposes.
- To automate the workflow using an ETL pipeline.
- To develop interactive Power BI dashboards for business users.
- To document the entire project and publish it on GitHub.

4. TECHNOLOGY STACK

TECHNOLOGY	PURPOSE
Databricks SQL	Data ingestion, transformation, and analytics platform.
SQL	Querying, cleaning, aggregation, and business analysis.
Delta Tables	Reliable storage layer supporting ACID transactions for Bronze, Silver, and Gold.
Power BI	Interactive dashboards and business intelligence visualization.
GitHub	Version control and project asset documentation.
CSV Dataset	Source product information data for the pipeline.

Core Architecture Components

Databricks SQL

Databricks SQL is used as the primary data engineering platform. It enables efficient SQL-based data processing, supports Delta Lake, and simplifies ETL workflow development.

SQL (Structured Query Language)

SQL is used extensively throughout the lifecycle to load data, clean datasets, remove duplicates, perform aggregations, build analytical views, and validate data quality standards.

Delta Tables

Delta Tables provide reliable and scalable storage by supporting ACID transactions, schema enforcement, and efficient querying layout.

Power BI & GitHub

Power BI connects seamlessly to the final Gold layer to display critical key metrics, whereas GitHub houses the repository assets, scripts, and document tracking.

5. DATASET DESCRIPTION

The project uses an Amazon Product Dataset containing information about products listed on Amazon across multiple categories, including pricing, discounts, ratings, customer review counts, and descriptions.

COLUMN NAME	DESCRIPTION
product_name	Name of the product listing
category	Product category classifications
actual_price	Original product manufacturer retail price
discounted_price	Discounted actual selling price on the platform
discount_percentage	Percentage of discount applied
rating	Average customer rating score
rating_count	Number of customer ratings submitted
about_product	Textual description overview of the product

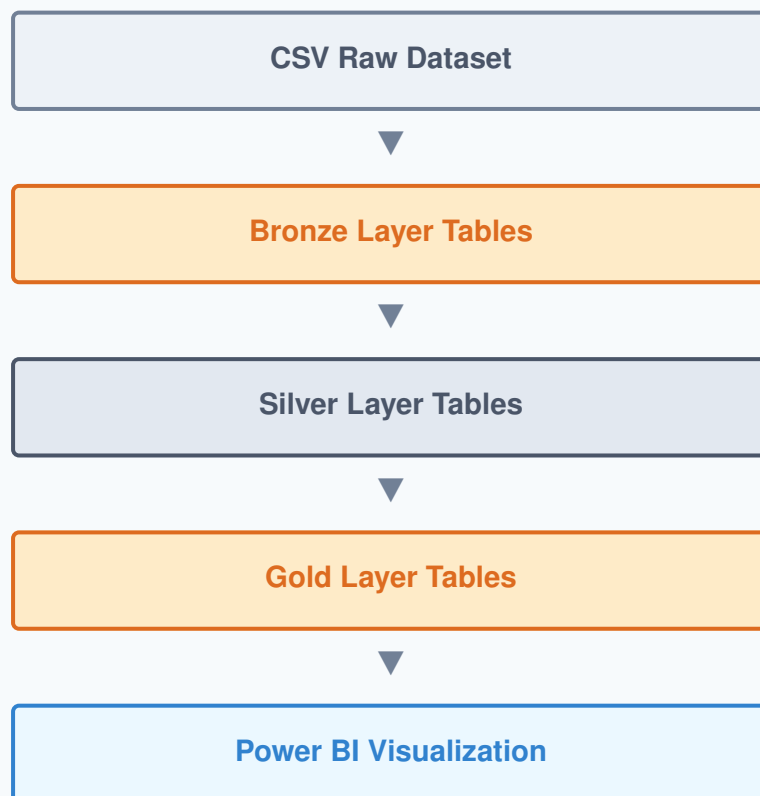


Chart 1: Data Lifecycle Pipeline Flow

6. PROJECT ARCHITECTURE

The project follows a modern layered lakehouse architecture that strictly separates raw data from processed, business-ready metrics. This prevents unintended mutations of historical facts and ensures pipeline re-runability.



7. MEDALLION ARCHITECTURE

Bronze Layer (Raw)

Initial landing zone. Stores unstructured/semi-structured data exactly as received from origin without modification, ensuring historical data preservation and audit traceability.

Silver Layer (Enriched)

Applies cleansing schemas, removes duplicate lines, enforces correct typing, parses price strings to floats, normalizes text representations, and strips whitespace formatting defects.

Gold Layer (Analytical Summary)

Contains data models aggregated and structured precisely to facilitate instant dashboard performance. Tables here map directly to key executive visualizations.

8. DATA INGESTION & 9. DATA CLEANING

Data Ingestion

Data ingestion acts as the foundational orchestration element. The raw file is moved into Databricks catalog storage and assigned a basic variant schema as a Delta Table view inside the `amazon_bronze` layer container.

Data Cleaning Operations

In the Silver layer phase, operations are applied sequentially to guarantee quality benchmarks:

- **Duplicate Line Removal:** Deduplication of metrics using item tracking keys.
- **Null Manipulation:** Dropping or applying defaults to text listings missing core data fields.
- **Type Normalization:** Stripping dollar/currency markers and casting text fields to numerical floats or integers.
- **Feature Engineering:** Constructing dynamic markers and metrics directly helpful for filtering buckets.

Strategic Benefits of Cleaning Phase:

- Drastically reduces query response layout times.
- Prevents skewed visual reporting anomalies.
- Supports reliable pattern detection without formatting noise.

10. EXPLORATORY DATA ANALYSIS & 11. BUSINESS SQL

Exploratory Data Analysis (EDA)

EDA provides insight into distributions, category product volume metrics, density checks, and ranking spreads before generating production reporting tables.

Business SQL Analysis

To deliver real-world business intelligence value, specific relational logic scripts were engineered to address executive inquiries:

1. Identification of Top Performance Categories

Determines product volumes per category to optimize product mix adjustments, inventory tracking, and stock balancing metrics.

2. High-Engagement Market Capture Analysis

Analyzes `rating_count` column distributions to uncover absolute consumer engagement rates, establishing consumer trust baselines.

3. Dynamic Pricing & Discount Assessment

Maps real selling spreads against standard manufacturing list rates to understand whether steep promotional drops drive client conversion or trigger low sentiment scores.

12. WINDOW FUNCTIONS & 13. ADVANCED SQL

Analytical window operations allow rows to retain distinct identities while computing relative partition ranks and rolling data points.

Core Functions Implemented

- `ROW_NUMBER()`: Allocates a sequential increment across category partitions to flag distinct top pricing.
- `RANK()` & `DENSE_RANK()`: Manages item prioritization scenarios containing tie conditions without leaving gaps.
- `LAG()` & `LEAD()`: Supports quick side-by-side transaction record evaluations.

Sample Analytics Source Snippet

```
SELECT product_name, actual_price,  
       DENSE_RANK() OVER (  
         PARTITION BY category  
         ORDER BY actual_price DESC  
       ) AS category_price_rank  
FROM amazon_silver;
```

Advanced Structural Components

Using Common Table Expressions (CTEs) breaks down code layouts into logical modular phases. This prevents duplicate evaluation steps and improves runtime execution layout speeds.

14. GOLD LAYER ANALYTICS DEFINITION

The business-focused analytical layer houses curated models optimized for interactive reporting. Rather than parsing thousands of raw events at runtime, Power BI polls these pre-compiled assets directly.

TABLE ASSET TARGET	ANALYTICAL FOCUS / VISUAL APPLICATION
gold_dashboard_kpi	Tracks macro metrics (Overall Count, Avg Rating, Avg Price).
gold_category_summary	Maintains grouped category statistics for comparative metrics.
gold_product_ranking	Preserves deep tiering matrices based on performance.
gold_discount_analysis	Stores markdown allocations and marketing distribution charts.
gold_price_segment	Casts items into Budget, Mid-Range, or Premium groups.

Performance Multiplier: By transitioning dashboards from raw tables to Gold summaries, network throughput and visual rendering delays dropped significantly.

15. DATA VALIDATION & 16. ETL LOGGING

Quality Control Validation Rules

Automated confirmation checks prevent bad data from reaching reporting components. Pipelines stop automatically if core criteria are breached:

```
-- Validation Check: Confirm logical selling bounds
SELECT COUNT(*) FROM amazon_silver
WHERE discounted_price > actual_price;

-- Validation Check: Enforce rating range limits
SELECT COUNT(*) FROM amazon_silver
WHERE rating < 0 OR rating > 5;
```

ETL Execution Logging Framework

Operational metadata tables monitor ingestion jobs, capture data volumes, and log processing speeds to assist with troubleshooting.

PIPELINE NAME	STATUS	RECORDS PROCESSED	EXECUTION DURATION
Amazon_Product_ETL	SUCCESS	1,465 Rows	3.0 Seconds

17. AUTOMATED ETL PIPELINE ORCHESTRATION

The modular layout steps are orchestrated into a repeatable sequence. This structure enables clear lineage tracking and allows specific components to be re-run independently if a step fails.

STAGE	ACTION FOCUS	OPERATIONAL PURPOSE
Step 1	Ingestion Engine	Loads source CSV data directly into the Bronze Layer without alteration.
Step 2	Data Scrubbing	Removes duplicates, standardizes strings, and handles missing values.
Step 3	Gold Materialization	Computes aggregates and stores them as reporting tables.
Step 4	Integrity Validation	Evaluates data bounds and logs execution metadata metrics.
Step 5	Reporting Sync	Refreshes dashboard views to display current system metrics.

18. BUSINESS INTELLIGENCE POWER BI DASHBOARDS

The curated Gold layer views serve as the direct foundation for a multi-page Power BI reporting application designed for business users.

| Primary Reporting Perspectives

1. Executive Overview Interface

Provides a high-level summary of operational metrics using KPI cards, overall distribution tracking, and category breakdowns.

2. Product Deep-Dive Interface

Enables clear tracking of item-level data points, performance trends, and discount distributions using scatter charts and dynamic lists.

3. Category Performance Analysis

Provides deep category-level insights into operational counts and rating trends to support product mix planning.

Key Interactive Visual Features:

- Cross-filtering layouts across metrics.
- Deep-dive navigation into specific categories.
- Responsive visual filters for real-time segmentation.

19. STRATEGIC BUSINESS INSIGHTS & 20. CHALLENGES

Analytical Insights Generated

- **Quality Drive Conversion:** Strong consumer sentiment metrics correspond directly to higher review volumes, highlighting customer satisfaction as a primary visibility driver.
- **Marginal Return on Discounts:** Deep promotional price markdowns do not guarantee a shift toward positive review metrics, proving product quality remains paramount.
- **Premium Resilience:** Higher-priced premium tiers consistently maintain strong customer satisfaction ratings, showing low price sensitivity among core buyers.

Challenges Addressed

Developing this pipeline highlighted several data engineering challenges:

- Parsing dirty source strings into structured numerical data.
- Structuring optimized data summaries to ensure fast dashboard performance.
- Developing a robust quality assurance strategy to catch formatting errors early.

21. FUTURE ENHANCEMENTS & 22. CONCLUSION

Future Enhancements Roadmap

- Automate data orchestration using Databricks Workflow Jobs.
- Transition from full batch updates to efficient incremental loading patterns.
- Build recommendation engines using native machine learning libraries.
- Implement automated Slack/Email operational alert notifications.

Conclusion

This project successfully demonstrates a complete end-to-end Data Engineering solution built on Databricks SQL and Power BI, following the Medallion Architecture.

By transforming raw data into structured, business-ready models, this solution delivers reliable insights that enable organizations to optimize product offerings and make data-driven decisions.

References

1. Databricks Workspace SQL Engine Guide: <https://docs.databricks.com/>
2. Microsoft Power BI Desktop Layouts: <https://learn.microsoft.com/power-bi/>
3. Delta Lake Storage Architecture Principles: <https://docs.delta.io/>